Homework 07

Use either R or Python.

Question 1: LDA

Let's start by estimating the components of LDA. From the lecture notes, for each target point $\mathbf{x}_0$, we find the class label k with the largest score

$$\mathbf{x}_0^\intercal \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^\intercal \Sigma^{-1} \mu_k + \log \pi_k.$$

Thus, we need to estimate:

- the prior probabilities π_k;
- the mean vector (centroid) μ_k for each class; and
- the pooled covariance matrix Σ.

Use saheart.csv, the `SAheart` data from `ElemStatLearn`. The two classes are defined by `chd`, the coronary heart disease indicator. Use all nine other variables as numerical predictors, coding `famhist` as 1 for `Absent` and 2 for `Present`. The following code constructs an $n \times 9$ predictor matrix $\mathbf{X}$ and an n-entry response vector $\mathbf{y}$, where n is the number of observations.

R

```r
heart <- read.csv("data/saheart.csv")
heart$famhist <- factor(heart$famhist, levels = c("Absent", "Present"))
X <- data.matrix(heart[, names(heart) != "chd"])
y <- heart$chd
```

Python

```python
import pandas as pd

heart = pd.read_csv("data/saheart.csv")
heart["famhist"] = heart["famhist"].map({"Absent": 1, "Present": 2})
X = heart.drop(columns="chd").to_numpy()
y = heart["chd"].to_numpy()
```

a. Calculate the prior probabilities, class means, and pooled covariance matrix from these data. Use the pooled covariance denominator $n - K$ from the lecture, where $K = 2$ is the number of classes.

b. Use your estimated values to predict the label of each observation in the training data. These are in-sample fitted labels. Provide the confusion matrix, with observed classes in rows and predicted classes in columns. Check that the dimensions of your matrices and vectors match. Calculate the scores yourself for this part.

c. Perform the same LDA analysis using `MASS::lda` in R or `LinearDiscriminantAnalysis(solver="svd")` in Python, using the same numerical predictors and empirical class priors. Provide the confusion matrix. Do the results match?

Answer

Part a

```
Class priors:
0    0.6537
1    0.3463
dtype: float64

Class means:
        sbp  tobacco     ldl  adiposity  famhist    typea  obesity  alcohol      age
0  135.4603   2.6347  4.3442    23.9691   1.3179  52.3675  25.7375  15.9314  38.8543
1  143.7375   5.5249  5.4879    28.1202   1.6000  54.4938  26.6229  19.1452  50.2938
```

```
Pooled covariance matrix:
                sbp  tobacco    ldl  adiposity  famhist   typea  obesity  alcohol      age
sbp         405.435   14.585  4.581     49.164    0.337 -15.588   18.939   64.400   95.134
tobacco      14.585   19.243  0.763      7.538    0.016  -2.057    1.833   20.517   22.765
ldl           4.581    0.763  4.001      6.033    0.092   0.345    2.660   -2.533    6.479
adiposity    49.164    7.538  6.033     56.753    0.433  -5.310   22.708   16.119   60.509
famhist       0.337    0.016  0.092      0.433    0.226   0.081    0.184    0.768    0.997
typea       -15.588   -2.057  0.345     -5.310    0.081  95.566    2.640    7.960  -20.278
obesity      18.939    1.833  2.660     22.708    0.184   2.640   17.615    4.689   15.697
alcohol      64.400   20.517 -2.533     16.119    0.768   7.960    4.689  598.277   27.886
age          95.134   22.765  6.479     60.509    0.997 -20.278   15.697   27.886  184.132

Dimensions:
X: (462, 9)
y: (462,)
Priors: (2,)
Means: (2, 9)
Sigma: (9, 9)
```

Part b

```
Manual LDA confusion matrix:
[[258  44]
 [ 73  87]]

Scores shape: (462, 2)
Predictions shape: (462,)
Training accuracy: 0.7467532467532467
```

Part c

```
Sklearn LDA confusion matrix:
[[258  44]
 [ 73  87]]

Confusion matrices match: True
All predictions match: True
```

The LDA implementation using `LinearDiscriminantAnalysis(solver="svd")` produced the same confusion matrix as the manually calculated LDA scores in Part (b): $\begin{pmatrix} 258 & 44 \\ 73 & 87 \end{pmatrix}$. Both methods correctly classified 345 out of 462 observations, giving an in-sample accuracy of 74.68%. All predicted labels matched, confirming that the manual implementation agrees with sklearn's LDA.

Question 2: QDA and marginal screening

In the lecture, ordinary QDA could not be fitted to the handwritten digit sample because its within-class covariance matrices were singular. Let's consider selecting fewer variables and see whether this helps. For this question, use the full zip-train.csv.gz as the training data and zip-test.csv.gz as the test data, with all ten digit classes. These are the original `zip.train` and `zip.test` data from `ElemStatLearn`. Each file has no header: the first column is the digit label and the remaining 256 columns are pixel values.

Instead of using all 256 predictors, select the 40 with the largest marginal variances in the training data. Here, marginal variance means the variance of one pixel across all training observations. Use the original pixel values without standardizing them, and retain the same selected predictors in the test data.

Use `MASS::qda` in R or `QuadraticDiscriminantAnalysis(tol=1e-8)` in Python.

a. Fit QDA using these 40 predictors and report the test confusion matrix, with observed classes in rows and predicted classes in columns. Also report the test misclassification rate.

b. Does the method work? Why do you think it works or does not work?

c. Repeat the analysis using the 10 predictors with the largest marginal variances in the training data. How does the test misclassification rate compare with the 40-predictor version? Why do you think this happens?

Answer

Part a

```
Training shape: (7291, 40)
Test shape: (2007, 40)

40-predictor confusion matrix:
     0    1    2    3    4    5    6    7    8    9
0  319    0   24    6    2    2    1    0    5    0
1    0  238    5    0   11    0    7    0    3    0
2    8    0  168    8    8    3    0    0    3    0
3    5    0    4  128    0   14    0    0   15    0
4    0    2   13    1  150    1    0    1    7   25
5   12    0    3    5    1  131    0    0    6    2
6    0    0    8    0    1    8  148    0    5    0
7    1    0    6    1    6    0    0  124    2    7
8    1    0   10   12    3    2    0    0  137    1
9    0    0    1    2    7    1    1    4    9  152

Misclassification rate: 0.1554559043348281
Number of errors: 312
QDA warnings: []
```

Part b

Yes, QDA works with the 40 selected predictors because reducing the number of variables makes the within-class covariance matrices nonsingular (full rank), so they can be inverted. Each digit class has enough observations to estimate its 40×40 covariance matrix. Unlike the original 256-predictor model, QDA can now calculate the discriminant scores and classify observations successfully, achieving a test misclassification rate of 15.55%.

Part c

```
10-predictor confusion matrix:
     0    1   2    3   4   5   6    7   8    9
0  298    0  11    1   1  16  28    4   0    0
1    0  241  10    0   3   1   1    0   2    6
2   19    2  95   14  14   3  22   21   5    3
3    5    0   6  103   3  42   0    0   7    0
4    1    5  14    0  95   2   4   15  17   47
5   15    0   6   25   4  96   4    2   8    0
6   41    0  15    4   0  18  78    7   5    2
7    4    1   6    0   6   0   1  122   0    7
8    1    2  11   36  12  11   4    1  74   14
9    0    2   9    2  20   0   0   16  10  118

10-predictor misclassification rate: 0.3423019431988042
40-predictor misclassification rate: 0.1554559043348281
```

Using the 10 predictors with the largest marginal variances, the test misclassification rate increased to **34.23%**, compared to **15.55%** with 40 predictors. This happens because selecting only 10 pixels removes important information needed to distinguish between digits. Although fewer predictors simplify covariance estimation, the loss of useful information reduces classification accuracy. Therefore, the 40-predictor QDA model performs better.

Question 3: Compare LDA and QDA

In this simulation, we will study how the relative performance of LDA and QDA changes with the true covariance structure. The parameter ρ controls the distribution from which we generate the data; it is not a tuning parameter of LDA or QDA.

Consider a classification problem with two equally likely classes, labeled 1 and 2, and $p = 10$ covariates. The first two covariates have the class-conditional distributions

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \mid Y = 1 \sim N_2\left(\begin{pmatrix} -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right),$$

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \mid Y = 2 \sim N_2\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}\right).$$

In both classes, the remaining eight covariates are independent standard normal variables, independent of the first two. Consider

$$\rho \in \{0, 0.2, 0.4, 0.6, 0.8, 0.9\}.$$

For each value of ρ, generate 100 training observations and 1,000 independent test observations from each class. Fit ordinary LDA and QDA using all 10 covariates and equal class priors. Estimate the class means from the training data, with one pooled covariance matrix for LDA and one full covariance matrix per class for QDA. Set a random seed and report it.

a. Generate one dataset for each value of ρ. Fit both methods and calculate their test misclassification rates. Within each setting, use the same training and test observations for the two methods.

b. At each fixed value of ρ, repeat the experiment 50 times, generating new training and test observations in each repetition. Plot average test error against ρ in one figure, with one curve for LDA and one for QDA. These curves compare performance across different data-generating distributions.

c. Compare the two methods at small and large values of ρ. How does changing the true covariance structure affect their relative performance? Explain the pattern using their covariance assumptions and the bias-variance trade-off. How can the eight noise covariates affect performance even though they contain no information about the class label?

Answer

Part a

```
Part (a): Test misclassification rates
 rho  LDA error  QDA error
 0.0     0.1505     0.1695
 0.2     0.1630     0.1880
 0.4     0.1510     0.1630
 0.6     0.1825     0.1785
 0.8     0.1710     0.1405
 0.9     0.1680     0.0935
```

Part b

```
Part (b): Average test errors
 rho  LDA error  QDA error
 0.0     0.1723     0.1980
 0.2     0.1714     0.1977
 0.4     0.1696     0.1903
 0.6     0.1682     0.1758
 0.8     0.1676     0.1353
 0.9     0.1705     0.1009
```

Part c

At small values of ρ, LDA generally performs better because the class covariance matrices are similar, making its shared covariance assumption appropriate. LDA estimates fewer parameters and therefore has lower variance. As ρ increases, the covariance matrices become more different, and QDA can perform better because it estimates separate covariance matrices for each class, reducing model bias.

However, QDA has higher estimation variance because it estimates more parameters. The eight noise covariates can worsen performance by introducing random variation into the estimated means and covariance matrices, even though they contain no useful information for classification. This illustrates the bias-variance trade-off between LDA and QDA.

Question 4: Create and use an LDA/QDA analysis skill

In Homework 5, you created a skill for formatting plots. Now create a reusable skill that guides an AI agent through an LDA/QDA analysis and produces a concise statistical report. You may work with AI to design and write the skill; you do not need to write every line yourself.

a. **Define the skill.** Write a single `SKILL.md` file no larger than **5 KB (5,000 bytes)**. Define the agent's behavior, analysis guidelines, and expected report structure. Keep the skill independent of this question: provide the dataset, response, and predictors separately when invoking it. The skill may guide the agent to explore lower-dimensional representations, such as principal component analysis (PCA), or to blend the class probabilities predicted by LDA and QDA. Keep all classifiers based on LDA, QDA, or their combination. Configure the skill to load **only when you explicitly invoke it**. Use your AI tool's invocation settings, or keep the file outside automatically loaded skill folders and supply it only in an explicit request.

Your skill must require a clearly readable analysis report of **at most one printed page**, containing a figure, a table, and numerical model-fitting results. You may specify the type and style of figures and tables you want the AI to produce. All instructions in the skill must be independent of the actual dataset, its variable names, and its observed results. The report should briefly describe the candidate models, the cross-validation procedure and results, the selected model, and its estimated prediction error. It should explain the main findings and any limitations.

b. **Apply the skill.** Explicitly invoke your skill to analyze the `SAheart` data from Question 1, using the same response and predictor coding. Ask the agent to seek the best predictive performance within the LDA/QDA methods allowed by your skill. Include ordinary LDA and QDA as comparisons; the extensions above are optional. Reserve a test set before choosing models. Use the same cross-validation folds within the training data to compare candidates and select any settings. Estimate any preprocessing, including an embedding, using only the training observations in each fold. Refit the selected model and preprocessing on all training observations, then evaluate the fitted classifier once on the reserved test set. Use misclassification rate to compare performance, and report your random seed.

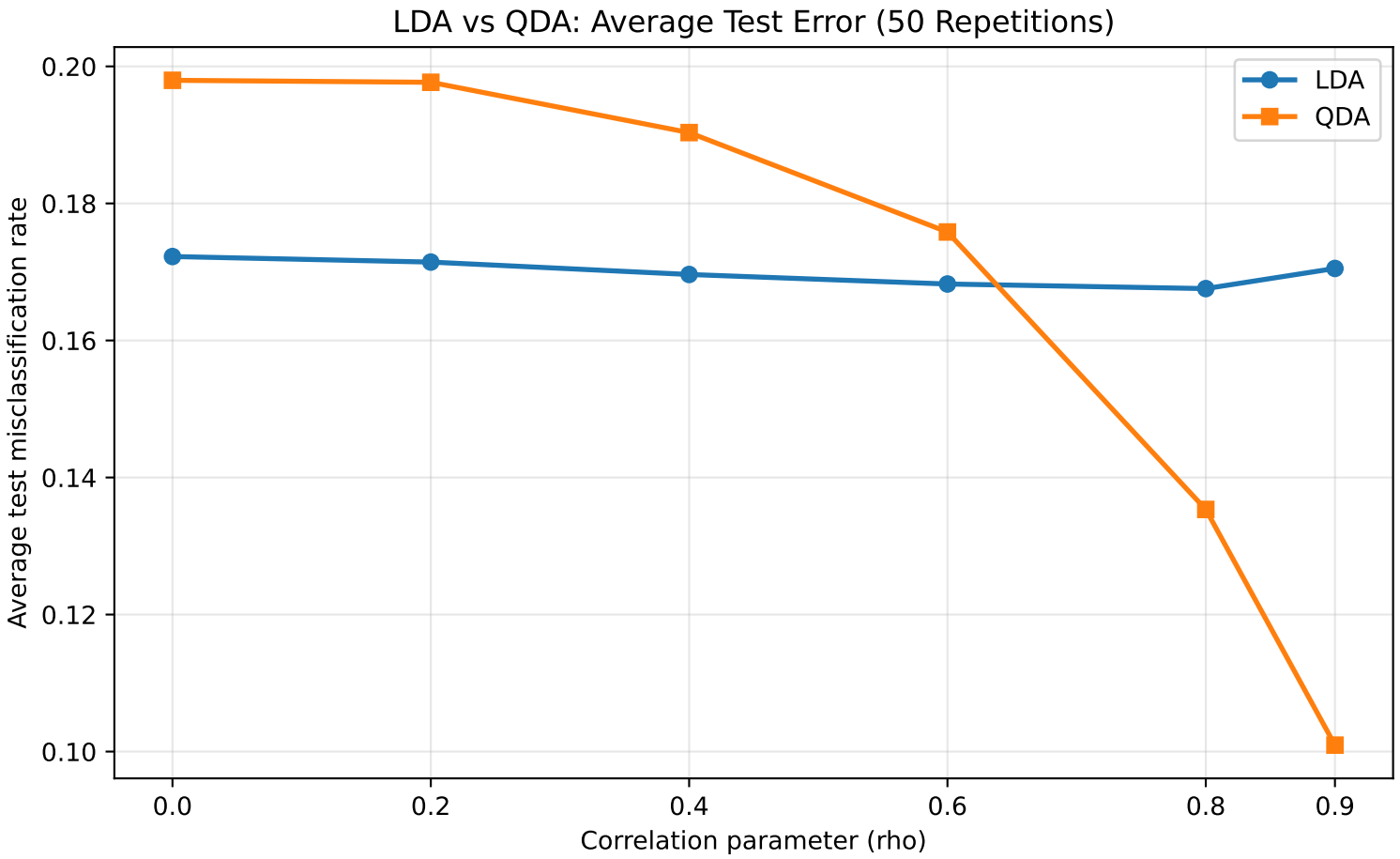


Figure 1: Average test misclassification rate of LDA and QDA over 50 repetitions at each ρ (seed 432).

c. **Review and submit.** Check the agent's analysis and make any skill revisions using only training and cross-validation results. Append the final one-page analysis report and the complete contents of your `SKILL.md` to your homework report. State the file size and briefly explain how you enabled explicit invocation, including the prompt used to apply the skill. The one-page limit applies to the analysis report; the appended skill and this explanation are outside that limit.

Your skill and analysis report will be reviewed and graded by other students through peer evaluation.

Answer

Part a

The skill is saved as `SKILL.md` in the user-level Claude Code skills folder (`~/.claude/skills/lda-qda-analysis/SKILL.md`). Its complete contents are appended in Part c.

Part b

The one-page analysis report produced by the skill (code in `q4_lda_qda_analysis.py`) begins on the next page.

LDA/QDA Classification of Coronary Heart Disease (SAheart)

Data. 462 men from the South African Heart Disease study; response `chd` (160 cases, 302 controls); nine numeric predictors (`famhist` coded 1 = Absent, 2 = Present). No missing values. A stratified 30% test set (139 obs.) was reserved before any modeling; the remaining 323 training observations were used for all model comparison. **Random seed: 2026.**

Candidates and validation. All models were compared by 10-fold stratified cross-validation (CV), using the same folds for every candidate and misclassification rate at a 0.5 probability threshold. Standardization and PCA were fitted inside each training fold only. Candidates: ordinary LDA and QDA; PCA ($q = 1, \ldots, 8$ components) followed by LDA or QDA; regularized QDA (shrinkage $\lambda = 0.1, \ldots, 0.9$); and a blend $w\,\hat{p}_{\mathrm{LDA}} + (1-w)\,\hat{p}_{\mathrm{QDA}}$ ($w = 0.1, \ldots, 0.9$), 36 settings in total. Ties in mean CV error were to be broken in favor of the simpler model (LDA family first).

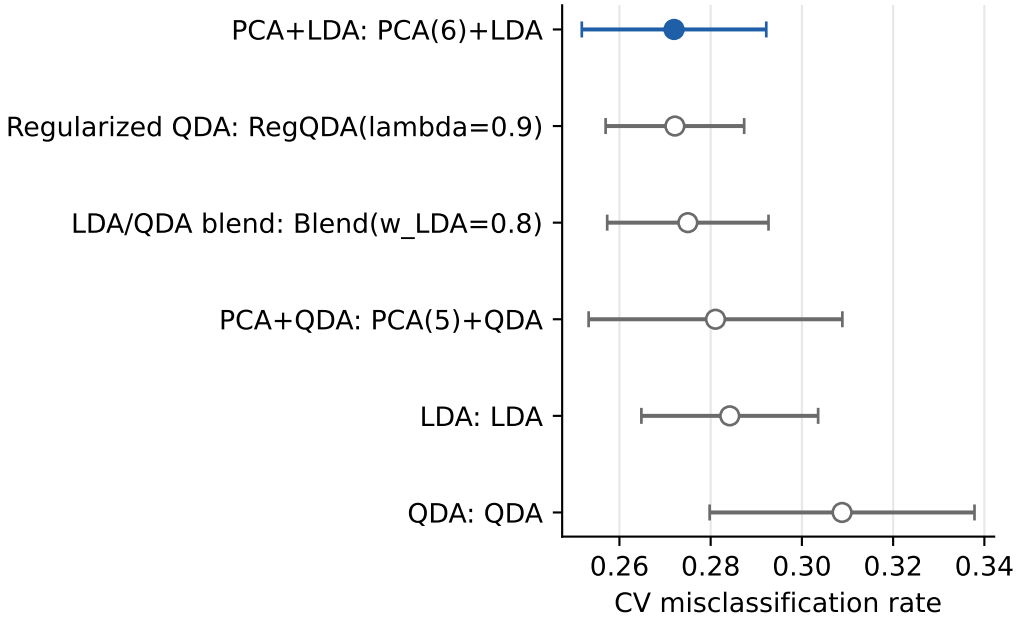


Figure 2: Mean 10-fold CV misclassification rate (±1 SE across folds) for the best setting in each model family. Filled marker: selected model.

Table 1: Best setting per family: mean CV misclassification rate and standard error (SE) over 10 folds.

Family	Best setting	CV error	SE
PCA+LDA	PCA(6)+LDA	0.272	0.020
Regularized QDA	RegQDA(lambda=0.9)	0.272	0.015
LDA/QDA blend	Blend(w_LDA=0.8)	0.275	0.018
PCA+QDA	PCA(5)+QDA	0.281	0.028
LDA	LDA	0.284	0.019
QDA	QDA	0.309	0.029

Selected model and test error. The lowest mean CV error was **PCA(6)+LDA** (standardize → first 6 principal components → LDA), CV error 0.272. Refitted on all 323 training observations and evaluated once on the reserved test set, it misclassified **34 of 139 (test error 0.245, binomial SE ≈ 0.036)**: of 91 controls, 12 were predicted as cases; of 48 cases, 22 were missed. Ordinary LDA had CV error 0.284 and ordinary QDA 0.309. No numerical warnings occurred, and both class covariance matrices had full rank (9).

Findings. Linear or heavily regularized boundaries did best: PCA+LDA, regularized QDA with strong shrinkage, and LDA-weighted blends were all within about 0.003 CV error of each other, while ordinary QDA was worst. With about 110 cases in training, the extra covariance parameters in QDA add more variance than the bias they remove.

Limitations. Differences among the top candidates are much smaller than one CV standard error (≈ 0.02), so the choice of PCA(6)+LDA is not clearly better than LDA or regularized QDA. Choosing the minimum over 36 settings makes the CV error slightly optimistic. The test set is small (139 obs.), so the test error is uncertain (±0.07 at 2 SE). The 0.5 threshold favors the majority class and misses many cases; misclassification rate ignores the unequal cost of missed cases. The best shrinkage, $\lambda = 0.9$, lies at the edge of the grid.

Part c

Skill Review and Revisions

I reviewed the model comparison using only the training and cross-validation results. The top three candidates had CV errors within 0.003 of each other, much smaller than the approximate standard error of 0.02. This suggests that the selected model was not clearly better than ordinary LDA. I also observed that the best shrinkage value, 0.9, was at the boundary of the searched grid. I did not revise the skill because the existing instructions already covered model comparison, cross-validation, and reporting limitations.

Explicit Invocation

I saved the skill in the user-level skills folder with `disable-model-invocation: true`, ensuring that it runs only when explicitly invoked using `/lda-qda-analysis`.

Prompt Used

I initially invoked the skill by typing `/lda-qda-analysis` without additional details. I then confirmed the following suggested analysis prompt in a follow-up message, changing its seed from 432 to 2026 ("lets do 2026"):

> Analyze `D:\Projects\stat432-fall2026\homework\week-07\data\saheart.csv`. Response: `chd`. Predictors: the other nine variables, with `famhist` coded 1 = Absent, 2 = Present. Seek the best predictive performance among the allowed LDA/QDA methods, including ordinary LDA and QDA as comparisons. Use seed 432. Save the code and the one-page report in `homework/week-07/`.

The remaining requirements (stratified test split, identical CV folds, preprocessing within each training fold, and the one-page report) came from the skill itself.

The final `SKILL.md` was kept unchanged after reviewing the training and cross-validation results.

File size. `SKILL.md` is 2,507 bytes, under the 5,000-byte limit.

Complete contents of `SKILL.md`:

```
---
name: lda-qda-analysis
description: Reusable LDA/QDA classification analysis. Use only when explicitly invoked by the user. Never load or apply
 ↪  automatically.
disable-model-invocation: true
---

# LDA/QDA Analysis Skill

## Activation
Apply this skill only when explicitly requested.
The user must separately specify the dataset,
response variable, and predictor variables.

## Objective
Compare LDA/QDA classifiers using honest
out-of-sample evaluation and produce a concise,
readable statistical report.

## Data preparation
- Inspect sample size, class balance, missing
  values, predictor types, and dimensions.
- Encode categorical predictors as specified.
- Reserve a stratified test set before model
  selection; never use it for tuning.
- Fit all preprocessing using training data only.
- Within cross-validation, fit preprocessing
  separately in each training fold.

## Candidate models
- Always compare ordinary LDA and ordinary QDA.
- Optionally compare regularized QDA, PCA+LDA,
  PCA+QDA, and probability blends of LDA/QDA.
- Use only LDA, QDA, or combinations of them.
- Apply PCA only within a fitted pipeline.
- Check covariance rank and numerical warnings.
- Do not silently discard failed candidates.

## Model selection
- Use stratified cross-validation with identical
  folds for every candidate.
- Select models and hyperparameters using mean
  validation misclassification rate.
- Specify a reproducible random seed.
- Use a predefined tie-breaking rule.
- Do not tune based on the reserved test set.

## Final evaluation
```

```
- Refit the selected pipeline on all training data.
- Evaluate it once on the untouched test set.
- Report test misclassification rate and counts.
- Clearly distinguish CV error from test error.
- Discuss uncertainty and limitations.

## Required report: maximum one printed page
Produce a clearly readable, self-contained report
containing:

1. Title and dataset summary.
2. Candidate models, validation procedure,
   and random seed.
3. Compact table of CV misclassification rates.
4. One labeled figure comparing model performance.
5. Selected model and numerical fitting results.
6. Final held-out test misclassification rate.
7. Brief interpretation and limitations.

Use readable fonts, labeled axes, sensible
rounding, and restrained formatting.
Keep the figure and table legible on one page.
Do not invent results or report unrun analyses.

## Output
Provide the one-page report and reproducible code.
Keep the skill itself separate from the report.
```